

CS-4112: Introduction to Machine Learning

Probability

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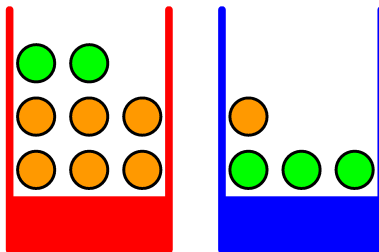
Slides by Dr. Nazar Khan

Probability Theory

- ▶ **Uncertainty** is a key concept in pattern recognition.
- ▶ Uncertainty arises due to
 - ▶ Noise on measurements.
 - ▶ Finite size of data sets.
- ▶ Uncertainty can be **quantified** via probability theory.

Probability

- ▶ $P(\text{event})$ is fraction of times event occurs out of total number of trials.
- ▶ $P = \lim_{N \rightarrow \infty} \frac{\# \text{successes}}{N}$.



$$P(B = b) = 0.6$$

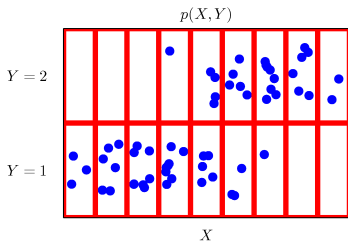
$$P(B = r) = 0.4$$

$$p(\text{apple}) = p(F = a) = ?$$

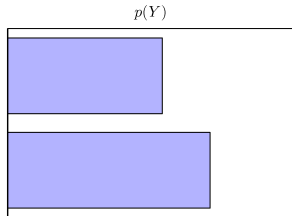
$$p(\text{blue box given that apple was selected}) = p(B = b | F = a) = ?$$

Terminology

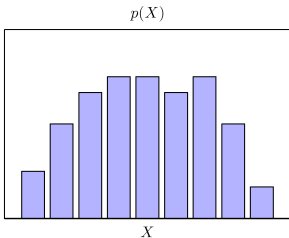
Joint Probability



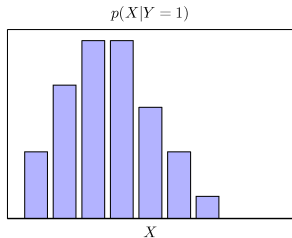
Marginal Probability



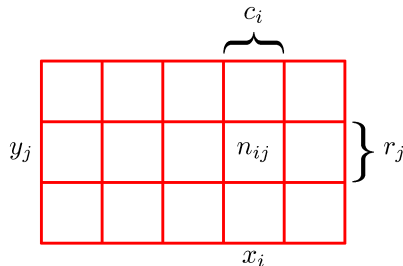
Marginal Probability



Conditional Probability



Elementary rules of probability



Elementary rules of probability

- ▶ Sum rule: $p(X) = \sum_Y p(X, Y)$
- ▶ Product rule: $p(X, Y) = p(Y|X)p(X)$

These two simple rules form the basis of [all](#) the probabilistic machinery that will be used in this course.

- ▶ The sum and product rules can be combined to write

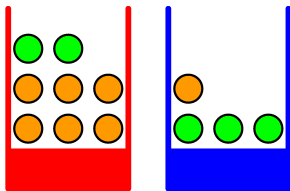
$$p(X) = \sum_Y p(X|Y)p(Y)$$

- ▶ A fancy name for this is **Theorem of Total Probability**.
- ▶ Since $p(X, Y) = p(Y, X)$, we can use the product rule to write another very simple rule

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$$

- ▶ Fancy name is **Bayes' Theorem**.
- ▶ Plays an *important role* in machine learning.

Terminology

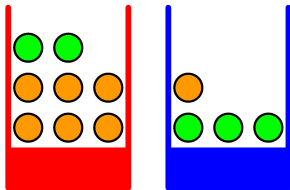


$$P(B = r) = 0.4$$

$$P(B = b) = 0.6$$

- ▶ If you don't know which fruit was selected, and I ask you which box was selected, what will your answer be?
 - ▶ The box with greater probability of being selected.
 - ▶ Blue box because $P(B = b) = 0.6$.
 - ▶ This probability is called the **prior probability**.
 - ▶ Prior because the data has not been observed yet.

Terminology



$$P(B = r) = 0.4$$

$$P(B = b) = 0.6$$

- ▶ Which box was chosen given that the selected fruit was orange?
 - ▶ The box with greater $p(B|F = o)$ (via Bayes' theorem).
 - ▶ Red box
 - ▶ This is called the **posterior probability**.
 - ▶ Posterior because the data has been observed.

Independence

- ▶ If random variable X is **independent** of random variable Y , then

$$P(X = x|Y = y) = P(X = x)$$

for all values x and y .

- ▶ Then, by the product rule

$$P(X, Y) = P(X|Y)P(Y) = P(X)P(Y)$$

- ▶ If joint $p(X = x, Y = y)$ equals the product of marginals $p(X = x)p(Y = y)$ *for all values x and y* , then random variables X and Y are **independent**.
- ▶ Intuitively, if Y is independent of X , then knowing X does not change the chances of Y and vice versa.
- ▶ Example: if fraction of apples and oranges is same in both boxes, then knowing which box was selected does not change the chance of selecting an apple.

Probability density

Continuous

- ▶ So far, our set of events was discrete.
- ▶ Probability can also be defined for continuous variables via

$$\text{Prob}(x \in (a, b)) = \int_a^b p(x) dx$$

- ▶ *Probability density function* $p(x)$
 - ▶ is always non-negative, and
 - ▶ integrates to 1.

Caution: Probability density is not the same as probability. Density can be greater than 1.

Probability density

Continuous

- ▶ Sum rule: $p(x) = \int p(x, y) dy$.
- ▶ Product rule: $p(x, y) = p(y|x)p(x)$
- ▶ Probability density can also be defined for a multivariate random variable $\mathbf{x} = (x_1, \dots, x_D)$.

$$p(\mathbf{x}) \geq 0$$
$$\int_{\mathbf{x}} p(\mathbf{x}) d\mathbf{x} = \int_{x_D} \cdots \int_{x_1} p(x_1, \dots, x_D) dx_1 \dots dx_D = 1$$